COSMOHACKATHON 2026

Evaluation criteria

Two-stage wildfire monitoring using remote sensing

Active Fire and Burn Severity
VIIRS, Sentinel-2 and Sentinel-1

Maximum: 100 points
48 points measured automatically + 52 points assessed by experts

Krasnoyarsk, 18-20 September 2026

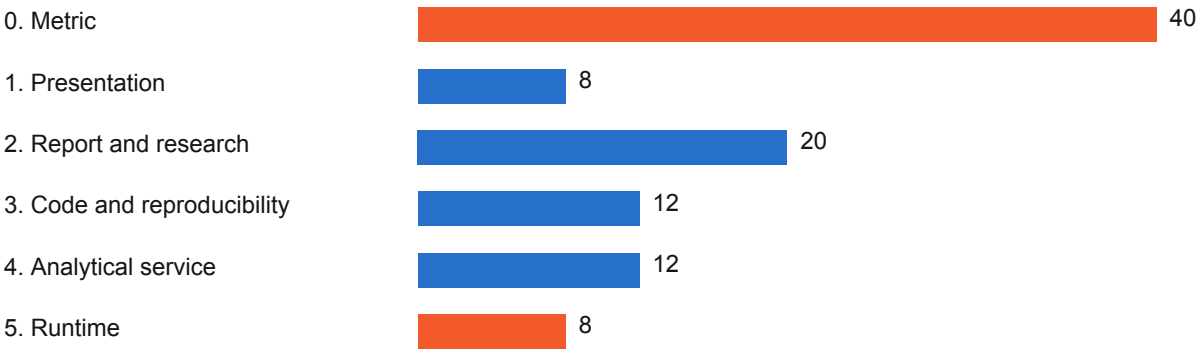
English translation of the original challenge document.
Original Russian source is preserved in repository history.

Case evaluation criteria: two-stage monitoring of wildfires according to remote sensing data

The final rating is considered by private selection. Maximum amount - 100 points. Sections 0 and 5 Measured automatically by the organizers, sections 1–4 are evaluated by experts in presentation and protection, report, repository and demonstration of information and analytical service.

A solution for which a valid submission.csv cannot be obtained receives 0 sectional 0 and on the criterion of "correctness of inferencing"; The remaining sections are evaluated in the usual manner.

48 points measured automatically; 52 assessed by experts



Picture 1. Distribution 100 sectional

SECTION maximum who appreciates	
0. Metrica 40 automatically, by private selection	
1. Presentation of the decision 8 defence expert	
2. Report and study 20 reporter	
3. Code quality, implementation, reproducibility	12 repository expert committee
4. Information and analytical service	12 demonstration panel
5. Speed of operation 8 automatically, on the infrastructure of the organizers	
Totally 100 —	

Section 0. Metric - before 40 scores

The best value of the Score composite metric obtained on a private sample is estimated. Submit, marked by the team as the final, and in its absence - the last technically valid.

```
Score = 0.35 * F1_af + 0.35 * IoU_burn + 0.30 * mIoU_sev
```

F1_af — F1Measurement of active combustion pixels on AF chips IoU_burn - IoU binary burn circuit on BS chips, mIoU_sev Average IoU in three classes of degree of damage. All components are considered micro-averaging over the total pool of pixels: TP, FP and FN are summed up across all chips, and only then are precision, recall, F taken1 and IoU.

Translation of the metric into points - linear normalization according to the results of all teams that have passed the formal test:

```
Ball = 40 * (Score_i - Score_worst) / (Score_best - Score_worst)
```

Score_best The highest value of a private metric among valid solutions Score_worst - the least, Score_i The value of the i-team metric. If a valid solution is one, it gets a complete solution 40 points.

The script for calculating the metrics is given to the teams together with the data: the organizers count according to the same code as the participants, there are no hidden amendments.

Sections 1–5. Expert assessment and speed measurement

Criteria for what balls are valued	
Section 1. Presentation of the decision — before 8 scores	
Clarity and readability of data and materials presented on slides. The logic of the presentation is traced, the key figures (metric values, increments relative to the baseline) are clearly given, the pipiline schemes are readable, the visualization of the results of detection and mapping are accompanied by explanations.	**0–3**
Confidence and public speaking skills, compliance with the rules. Ability to answer questions by solution: substantiation of the selected approaches, understanding of the subject specifics (sources of false positives, limitations of the index approach, choice of a before / after pair), correct interpretation of the obtained values of the metric.	**0–5**
Section 2. Report and study - before 20 scores	

Criteria for what balls are valued		
EDA and observations of data	A meaningful analysis of the data with conclusions that influenced the team's decisions: class imbalance and the proportion of burning pixels, the distribution of the degree of damage classes, differences between regions and types of land cover, the effect of cloudiness and the proportion of valid pixels, the nature of noise in radar channels. Measurable facts and a bundle of "observation" are expected → Decision taken." Graphs without conclusions are not considered a full EDA.	**0–4**
Baseline and starting point	The team described the issued baseline as a point of reference: the purpose, key components of the pipeline of both modules, strengths and weaknesses, identified limitations (fixed dNBR thresholds on a heterogeneous cover, lack of filtration of man-made thermoanomalies) and the hypotheses resulting from them. It should be clear what the team was improving and through what changes the quality was expected to increase.	**0–4**
Experiments and comparisons	The experimental part allows you to trace the progress of the work and check the validity of the final choice: the conditions of the experiments (validation scheme, partitioning, random initial values) are recorded, the results are compared according to a single metric, the options are compared correctly. Ablations for individual components and an explanation of the causes of the effect are welcome.	**0–4**
Engineering and Subject Solutions	Specific solutions are described,aimed at quality in this particular setting: work with a strong imbalance of classes, processing of clouds and shadows along the SCL layer, filtering of technogenic and reflective objects in the detection module, taking into account the type of land cover and relief when determining the degree of damage, using radar data to verify contours, resistance to the difference in phenological states between the dates "before" and "after". It should be clear how the mechanisms are built into the Pipeline and why they improve quality.	**0–4**
Conclusions and further development	Reasonable conclusions based on the results: what solutions gave the main contribution, what types of errors remain problematic (skipping small foci, false contours on removed fields, mixing neighboring classes of the degree of damage), what limitations the approach has on data, calculations and conditions of applicability, a short plan for further improvements. It is important to link the conclusions with the conducted experiments.	**0–4**
Section 3. Code quality, implementation, reproducibility - up to 12 scores		

Criteria for what balls are valued		
The Organizers can deploy and run the solution strictly according to the instructions providedNo manual code edits. README.md clearly specifies the requirements for the environment, the establishment of dependencies, commands to start inferencing and training, the expected structure of the input data, the procedure for preparing and connecting the scales. Instructions result in submission.csv in the required format. Dependencies are fixed (requirements.txt, pyproject.toml with lock-file or Dockerfile), random initial values are fixed, re-run gives the same metric value with no worse accuracy 0,005.		**0–5**
Project structure and readability	The code is organized logically: a clear structure of directories, separation of modules (data, model, training, inference, service, utilities), configurations and launch parameters are brought out, there is no redundant hard code and “magic” numbers without explanation. Critical are clear entry points, explanations where logic is not obvious, careful work with paths and I/O, the absence of absolute paths from the developer’s machine.	**0–4**
Correctness of inferencing	The interface is executed correctly and stably on a full test set: the script generates submission.csv strictly in the required format, a set of pairschip_id, class_id) completely matches the template, RLE is decoded without errors, indexes do not go beyond the chip, masks of different classes within the same chip do not intersect, there are no omissions and NaN, high-cloud chips are processed without falling.	0 or 3
Section 4. Information and analytical services – up to 12 scores		
Performance of the service	The service runs on the command described in README.md on the side of the organizers or is demonstrated by the team on defense, receives a space-time request (polygon or limiting rectangle and date interval) and returns the result for a finite time without manual intervention. Having a REST API that allows you to get the same result without a user interface is a must.	0 or 4
Cartographic inference and vector contours	The service displays a map of thermopoints for the specified period and the contours of the area covered by fire on top of the cartographic substrate, and also allows you to unload the contours in GeoJSON or Shapefile with the correct coordinate system and attributes: contour identifier, class of the degree of damage, area. The correctness of geometry, completeness of attributes and ease of unloading are evaluated.	**0–4**

Criteria for what balls are valued		
Analytical background	The service forms a certificate for a given territory and period: the total area of burns in hectares and the distribution of the area in three degrees of damage, with unloading in a machine-readable format. The correctness of the calculation of the area (taking into account the resolution and projection), the completeness and readability of the reference, the reproducibility of the numbers given in it according to the issued data are estimated.	**0–4**
	Section 5. Speed of operation - up to 8 scores	
Measurement is carried out on the infrastructure of the organizers from the launch of inferencing (entrance is a catalog with test chips and sample_submission.csv) until submission.csv is available in the required format, including weight loading and preprocessing.	8 Scores - the decision worked for less than 30 seconds; 6 Scores - the decision worked for no more than 5 minutes; 4 Score – the decision has worked for no more than 12 minutes; 2 Score – the decision has worked for no more than 30 minutes; 0 More than an hour or the decision didn’t work.	**0–8**

Calculation procedure

The procedure of actions of the organizers after the closing of decision-making:

The final submitt of each command is verified by a script. untested
receiver 0 sectional 0 and on the criterion of "correctness of inferencing";

- according to the past, a score on a private sample is considered;

Score values of all valid solutions are normalized into section points 0;

The inferencing time of each decision is measured and compared with the baseline time
decisions of the organizers;

- sections 1–4 Evaluated by experts independently, differences are removed
discussion;

- points are summed up, with equal amounts, a team with a large Score becomes higher.

Dispute cases on formal criteria are resolved in favor of the team, if the violation does not give an advantage in the metric and does not interfere with the reproduction of the decision.

Grounds for Disqualification

the use of data relating to the territories and dates of the test sample, including
finished products of active combustion and burnt areas;

Attempts to restore georeference or dates of test chips;

Manual marking of test chips by a person;

The inability to reproduce the declared result at the request of the organizers.